EXTREMAL GRAPH THEORY

Erdős Problem 915

From Proof to Search

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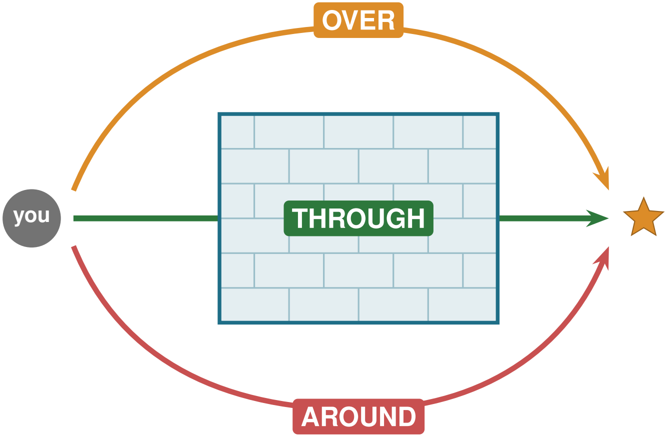
A hard problem is a wall



Three ways past the wall

THROUGH

Proof

one argument,
every n at once

OVER

Calculation

a finite check,
with a certificate

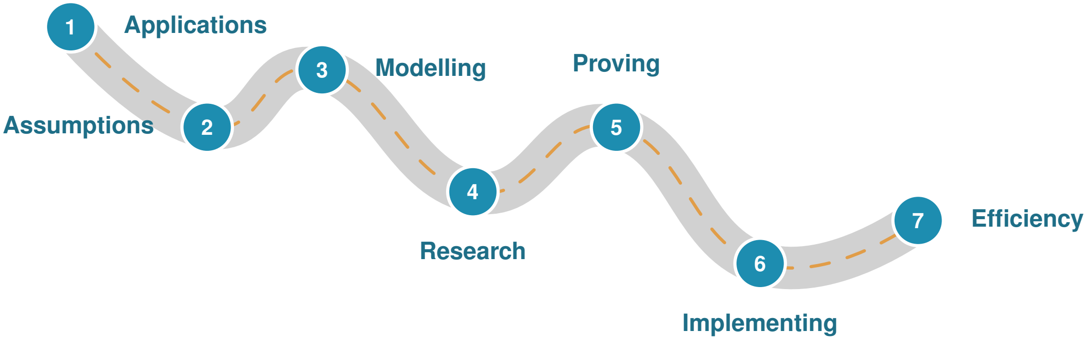
AROUND

Approximation

search for examples,
a lower bound

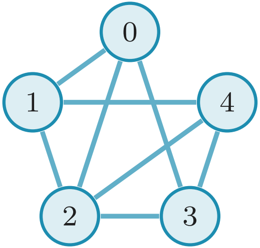
This thesis uses all three.

The road of a research problem



The slides ahead follow this road.

Problem 915, precisely



K_5 minus 2 edges: $8 = \ell_4(5)$ edges

A graph: n vertices, E edges.

Routes. $\lambda(u,v)$ = independent u-v paths.

Rule. No pair gets m routes: $\lambda^{\max} \le m-1$.

Goal. Pack in as many edges as possible.

$$\ell_m(n) = \left\lfloor \frac{m(n-1)}{2} \right\rfloor \quad \text{(Mader)}$$

What is it good for?

No direct application. The methods are the point.

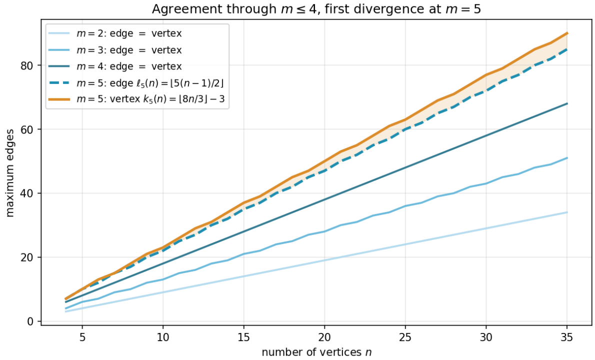
The question	The tools	Where it lives
network redundancy	flow, search, proofs	grids, pipelines, roads

One question, twelve variants

Model	simple	multigraph	hypergraph	$\times 3$	
Direction	undirected	directed		$\times 2$	$\boxed{= 12}$
Separation	edge-disjoint	vertex-disjoint		$\times 2$	

Mader's theorem is one corner. The other eleven are the journey.

Edge routes versus vertex routes



λ: edge-disjoint. κ: vertex-disjoint.

$m \le 4$ same answer (Leonard).

$m = 5$ first gap: $k_5(n) = \left\lfloor \frac{8n}{3} \right\rfloor - 3$

(Sørensen-Thomassen, $n \ge 13$).

$m \ge 6$ open since 1974.

What fell, and how hard

Proved by hand	Exact value conjectured	Leading term only	Open
undirected edge, all models	directed arc $m \geq 3$	directed hyper, arc	vertex-disjoint $m \geq 6$
multigraph vertex (convention)	directed vertex $m \geq 3$	directed hyper, vertex	hypergraph vertex $m \geq 4$
hypergraph vertex $m \leq 3$			
directed multigraph arc			
directed simple, $m = 2$			

easier → harder

How the machine works

Which search wins

variant	best	SA	tabu
simple undirected $n{=}7$	9	9	**9**
simple directed $n{=}7$	18	16	**18**
directed multi $n{=}7$	24	20	**24**

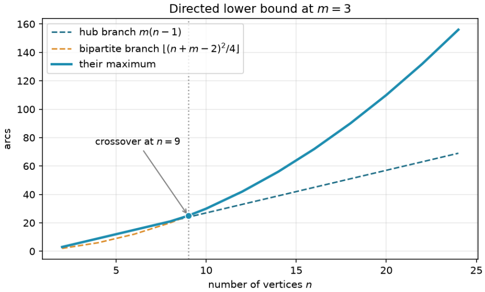
Tabu reaches the hard directed optima.

What makes it fast

C hot-path routine	speedup
dense max-flow	$2\times$
connectivity test	$2.2\times$
canonical form	$\mathbf{147}\times$

Python core, optional C, same answers.

New results: hypergraph vertex and directed arcs



The quadratic family overtakes near $n = 9$.

Four genuine advances

solved **Hypergraph vertex**, $m \le 3$

$$k_m^{(r)}(n) = \left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$$

solved **Directed multigraph**, all n and m

$$L_m^{\mathrm{dir}}(n) = (m-1)\max\left(2(n-1), \left\lfloor \frac{n^2}{4} \right\rfloor\right)$$

leading term **Directed simple**, both

$$\ell_m^{\mathrm{dir}}(n) = k_m^{\mathrm{dir}}(n) = \frac{n^2}{4} + \Theta_m(n)$$

leading term **Directed hypergraph**, both

$$(1 + o(1))\,\frac{(m-1)n^2}{4(r-1)}$$

The whole landscape

Separation	Model	Value	Status
Undirected			
edge	simple	$\lfloor m(n-1)/2 \rfloor$	proved (Mader)
edge	multi	$(m-1)(n-1)$	proved
edge	hyper	$\lfloor (m-1)(n-1)/(r-1) \rfloor$	proved (Gomory-Hu)
vertex	simple	$\lfloor m(n-1)/2 \rfloor$ $(m \le 4)$, $\lfloor 8n/3 \rfloor - 3$ $(m{=}5,\, n \ge 13)$	cited, open $m \ge 6$
vertex	multi	adjacency count, so the simple value	convention
vertex	hyper	$\lfloor (m-1)(n-1)/(r-1) \rfloor$ $(m \le 3)$	proved $m \le 3$, open $m \ge 4$
Directed			
arc	simple	$\max(2(n-1), \lfloor n^2/4 \rfloor)$ $(m{=}2)$, $n^2/4 + \Theta_m(n)$ (all m)	proved, conj. $m \ge 3$
arc	multi	$(m-1)\max(2(n-1), \lfloor n^2/4 \rfloor)$	proved, every n and m
arc	hyper	$(1+o(1))(m-1)n^2/(4(r-1))$	leading term, value open
vertex	simple	$\max(2(n-1), \lfloor n^2/4 \rfloor)$ $(m{=}2)$, $n^2/4 + \Theta_m(n)$ (all m)	proved separately, conj. $m \ge 3$
vertex	multi	adjacency count, so the simple digraph value	convention
vertex	hyper	$(1+o(1))(m-1)n^2/(4(r-1))$	leading term, value open

proved **leading term** **conjecture** **open**

What is left, and why it matters

Open problems

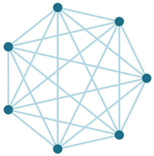
- the structural decomposition, of which the backward arc is the hardest quarter
- the linear coefficient, one-way, arc and vertex alike
- vertex-disjoint, $m \geq 6$
- hypergraph vertex, $m \geq 4$
- the exact directed hypergraph value at finite n

One pipeline, every variant

- one representation
- one exact checker
- one certifier
- one search

Thank you

Questions?

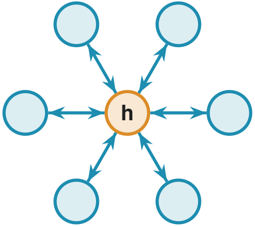


Backup slides follow.

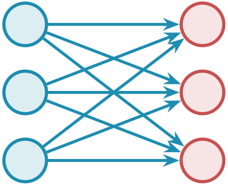
BACKUP SLIDES

Dive deeper into any single variant.

Backup: the directed arc conjecture



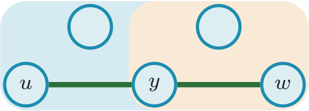
bidirected hub: $2(n-1)$



bipartite digraph: $\lfloor n^2/4 \rfloor$

At $m = 2$ the value $\max(2(n-1), \lfloor n^2/4 \rfloor)$ is proved (the two constructions tie at $n = 7$). For $m \geq 3$ the conjecture adds the augmented-bipartite family, and the leading term $n^2/4 + \Theta_m(n)$ is now proved unconditionally, so only the linear coefficient is at stake. Pinning it needs a **structural decomposition** $V = A \cup B$, complete forwards, empty inside A, nothing coming back, of which the backward-arc clause is the hardest quarter.

Backup: the hypergraph vertex problem



a Berge path u, e_1, y, e_2, w

Routes are **Berge paths**, alternating vertices and hyperedges.

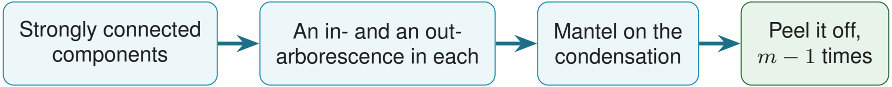
Solved ($m \leq 3$, every r, all n):

$$k_m^{(r)}(n) = \left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor,$$

via a rank bound on incidence graphs (Tutte SPQR, valid where $\kappa \leq 2$).

Open at $m \geq 4$: needs a 4-connectivity analogue.

Backup: peeling a directed multigraph apart



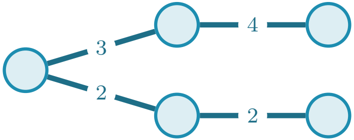
A **reachability skeleton** is a spanning subgraph that on its own reproduces every reachability relation the digraph has. Two facts close the problem:

(1) one always exists with at most $M(n) = \max(2(n-1), \lfloor n^2/4 \rfloor)$ arcs;

(2) deleting one lowers *every* pairwise connectivity by exactly one.

So $m-1$ peels account for every arc: $L_m^{\mathrm{dir}}(n) = (m-1)M(n)$, for every n and every m. No parity split, no auxiliary conjecture, no size limit, and $L_3^{\mathrm{dir}}(7) = 24$ falls out in three lines.

Backup: how connectivity is stored



A **Gomory-Hu tree** stores all $\binom{n}{2}$ connectivities in $n-1$ numbers: $\lambda(u,v)$ is the lightest edge on the tree path, and $\lambda^{\max}$ is the heaviest tree edge.

Backup: values at a glance

Object	Extremal value
simple undirected edge, $\ell_m(n)$	$\lfloor m(n-1)/2 \rfloor$
multigraph edge, $L_m(n)$	$(m-1)(n-1)$
hypergraph edge, $\ell_m^{(r)}(n)$	$\lfloor (m-1)(n-1)/(r-1) \rfloor$
simple vertex, $k_m(n)$ $(m \leq 4)$	$\lfloor m(n-1)/2 \rfloor$
simple vertex, $k_5(n)$ $(n \geq 13)$	$\lfloor 8n/3 \rfloor - 3$
hypergraph vertex, $m \leq 3$	$\lfloor (m-1)(n-1)/(r-1) \rfloor$
directed simple, $m = 2$ (arc and vertex)	$\max(2(n-1), \lfloor n^2/4 \rfloor)$
directed multigraph arc, every m	$(m-1)\max(2(n-1), \lfloor n^2/4 \rfloor)$
directed simple, every m (arc and vertex)	$n^2/4 + \Theta_m(n)$
directed hypergraph, both separations	$(1+o(1))(m-1)n^2/\big(4(r-1)\big)$
random-graph threshold	$p^* = m/n$